[COMPANY NAME REDACTED]

the Platform
System Analysis and Design Document

Logistics operations, operation approval, billing, accounting bridge and general ledger platform

Document	System Analysis and Design (SAD)
Version	0.1 (revised to the origin/main state)
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Status	Draft for review
Prepared by	Engineering
Owner	the Company management

[Company Name Redacted]
[Address Redacted]

Companion document: Software Requirements Specification (SRS) v0.1. Requirement identifiers used here (FR-, NFR-, IFR-, DR-) are defined in the SRS. Status labels: **Built** = implemented in the repository; **Partial** = implemented in part; **Planned** = not implemented (for example the credit note document and the revenue split, listed as missing in the demo known-gaps document).

Document Control

Revision History

Version	Date	Author	Description
0.1	November 8, 2026	Engineering	First issue, based on a stale local checkout.
0.1 (rev. 2)	November 8, 2026	Engineering	Re-based on origin/main. Adds operation approval, the accounting bridge states, the company-specific ledger, air consolidation (MAWB with many HAWBs), the demo seed, the public tracking lookup with proof-of-work, and the known-gaps list as Planned items.

Review and Approval

Role	Responsibility	Signature	Date
Business owner	Confirms business scope and process description		
Accounting lead	Confirms ledger, billing and posting rules		
Technical lead	Confirms architecture, deployment and security design		

Source of Information

This document was prepared from the project repository at origin/main: the web client, the API server, the database schema, the operation status and billing modules, the continuous integration and deployment workflows, the demo rehearsal notes (known gaps and demo data), the freight standards notes and the design and development reports. Where the repository does not show a capability, the capability is marked Planned. No production data, credentials or customer records are reproduced.

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1. Executive Summary

the Platform is the internal operations platform of the Company, a freight forwarder handling ocean and air freight (export and import), air consolidation under master air waybills, vehicle shipping on roll-on/roll-off vessels, domestic pickup and delivery cartage, and fleet dispatch. It records a shipment from quotation and booking, through an approval step, to billing, the accounting bridge and the general ledger, and it gives customers a public tracking lookup.

The platform is one web application. Logistics staff book files; a manager approves them; the billing engine rates approved files; an accountant verifies and posts the resulting items to the ledger of the company. Accountants see only approved files. Every mutating action is written to an audit trail.

Key design decisions

No.	Decision	Reason
D-01	One web application with a single sign-in for logistics and accounting	Staff move between freight files and the ledger in one task; one shell and one audit trail keep the record consistent.
D-02	Every operational record carries a company code and is filtered by it	The data model supports more than one company; records of each company stay separate.
D-03	Two SQLite databases: the application database and a separate accounting database	The ledger has its own lifecycle, can be backed up and moved on its own, and is isolated from freight data.
D-04	An operation approval step sits between booking and accounting	Billing is released only for approved files, and accountants see only approved files.
D-05	Billing lines come from a pure rating engine using tariffs, charge packages and accepted quotes	Rates can be checked in isolation and the same file gives the same charges.
D-06	Ledger postings go through an accounting bridge that stages, trial verifies and posts each item	Freight staff do not write to the ledger directly; the accountant decides when items post.
D-07	Schema changes in production are additive and applied at start-up	Production starts with a plain Node process and has no migration command.
D-08	Deployment to a single production slot from the main branch	One deployment path keeps production traceable to a reviewed commit.

2. Business Context and Analysis

2.1 Organizational need

the Company needs one controlled record of each shipment from quotation to the final ledger entry. Staff need one file number to link a booking, containers or vehicles, cartage orders, dispatch, tracking milestones, invoices and accounting entries. Managers need to approve a booking before it is billed. Customers need to see the status of their cargo without calling the office.

2.2 Problem statement

Freight, cartage and accounting work is spread across several people. Without a shared file number and an approval step, the same cargo can be billed twice, a cost can reach the ledger from two routes, a booking can be billed before it is checked, or a period can be closed before its entries are complete. The platform answers these with links between records, an approval state machine, one billing engine, a bridge that refuses items it cannot post, a period lock in the ledger and an audit trail.

> The repository does not contain measured figures for delays or losses in the previous way of working. This document states the design goals qualitatively. Measured figures for the platform itself come from the demo rehearsal and are quoted in Section 12.

2.3 Current state

The platform is at the state of the origin/main branch. Its demo rehearsal (5 November 2026) and the demo presentation script (6 November 2026) exercise every module with a seeded demo copy. The live site is hosted on Azure App Service and deploys automatically from the main branch. A legacy ledger cleanup command exists but has not been run on production (Section 13). A stopped earlier production instance remains a separate owner decision.

2.4 Objectives

Table 2-1. Business objectives and how the design meets them

Objective	Design response
One record per shipment	The file number is the reference every related record stores (vehicles, cartage, routes, tracking, invoices, bridge items).
Checked before billed	Operation statuses (draft, pending, for correction, approved) gate billing and accounting.
Correct charges	The billing engine applies the most specific tariff, honours accepted quotes and keeps manual lines unchanged.
Controlled posting	Bridge items are staged, trial posted, then posted; unmapped charge codes are refused with a reason; a posted invoice is reversed, not edited.
Closed periods stay closed	The ledger refuses new or re-dated entries in a closed month.
Traceable changes	Mutating actions write an audit record; booking status changes write an event with the user, time and note.
Customer visibility	A public lookup by file, booking, air waybill, master air waybill or house air waybill returns tracking status.

2.5 Scope

In scope

- Ocean and air freight files for export, import and domestic movements, including containers, dimensions, weights and volumes.
- Air consolidation: one master air waybill with many house air waybills, and per-house arrival notices.
- Operation approval of freight files.
- Vehicle inventory for roll-on/roll-off shipments; pickup and delivery cartage; fleet, drivers, dispatch and proof of delivery.
- Rates: directories, tariffs, billing codes, charge packages, quotes and commodity catalogue with HS codes.
- Billing lines, invoices, the billing pipeline, the accounting bridge and the ledger for the company.
- Tracking milestones, the public lookup and the live ship and aircraft map.
- Users, companies, roles, sign-in security, audit trail, attachments, support requests and a demo seed.

Out of scope

- Payroll, customer relationship management, and warehouse management beyond cargo location fields.
- Direct connection to government electronic filing and to carrier rate APIs (Planned; only status records exist).
- Customer sign-in to a customer portal (Planned; web account records exist).
- Bank file transfers. Checks are recorded and printed by staff.

2.6 Stakeholders and user roles

Table 2-2. Stakeholders and user roles

Role	Typical users	Access in summary
superadmin	System owner and technical administrator	All modules, users and companies, approvals and returns. Only a superadmin can change another superadmin account.
manager	Branch or department manager	All logistics and ledger functions; approves and returns files; no user administration.
operator	Freight, cartage and dispatch staff (encoders)	Creates and edits draft and returned files, submits them; no ledger access; cannot approve.
accountant	Accounting staff	Sees approved files only; billing, bridge and ledger; read-only on freight records.
viewer	Read-only staff and auditors	Read access; writes refused by the server.
Public visitor	Customer or any person with a reference	Public lookup only, each lookup behind a proof-of-work token. No account.
Customer web account	Customer with a web account record	Records hold permission flags. A customer sign-in screen is Planned.

3. Business Process Analysis

3.1 End-to-end freight lifecycle

A file passes through the steps of Figure 1. The file number links every step. Billing for a file is released only after approval, and the ledger is reached only through the bridge.

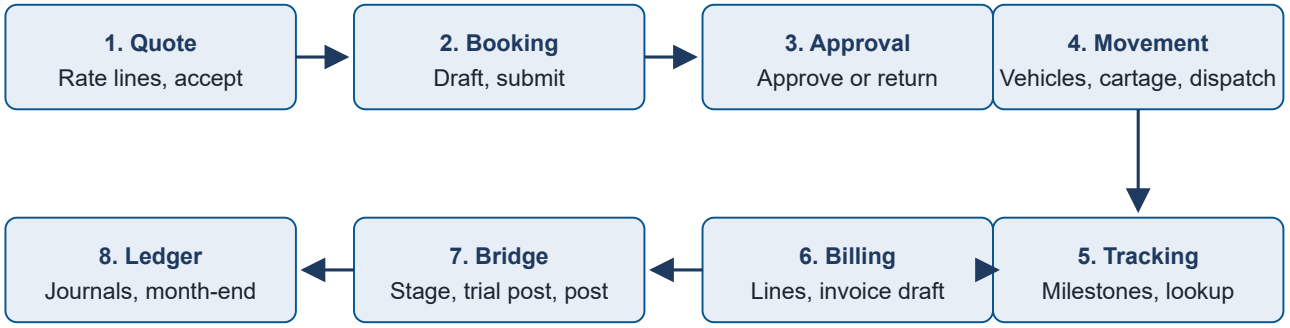


Figure 1. Freight lifecycle from quotation to ledger. The approval step (3) gates billing (6); the bridge (7) is the only route into the ledger.

3.2 Process steps

Table 3-1. Process steps, owner and system record

Step	Activity	Owner	System record and rule
1	Quote	Operator or manager	Quote with status Draft, Sent, Accepted, Rejected or Expired. An accepted quote can be converted into a file.
2	Booking	Operator	Freight file with a file number from the booking form (prefix by mode and direction). Required fields are checked before submit (shipper, consignee, bill-to, origin, destination, dates, cargo).
3	Approval	Operator submits; manager or superadmin approves or returns with a note	Operation status moves draft to pending to approved, or to for correction and back to pending. Each step writes an event.
4	Cargo and documents	Operator	Containers, air dimensions, HAWB rows for consolidations and printable documents (bill of lading, air waybill, manifest, packing list, invoice, export declaration).
5	Movement	Operator or dispatcher	Vehicle records (RORO), cartage orders, dispatch routes with driver, vehicle, stops and proof of delivery.
6	Billing	System on save; accountant reviews	Rating creates billing and cost lines; the invoice is drafted, issued, posted or voided. Released for approved files only.
7	Bridge	Accountant	Items are Staged, then Trial Verified (journal balances, checked without writing), then Posted.
8	Ledger and close	Accountant	Vouchers, journals and cash entries; month-end closes the period; reports. A posted invoice is reversed with a reversal voucher.

3.3 Approval design

Operation statuses are the approval step between operations and accounting. Operators and encoders (operator, manager, superadmin) create and edit draft and returned files and submit them. A pending file is locked for everyone. Only managers and superadmins approve or return; a return needs a note and may carry an image. An approved file is locked for operators; managers, superadmins and accountants may still edit it under their rules. Accountants see approved files only. Legacy rows that existed before the approval step are approved by default.

> The role check separates operators from approvers. The code does not stop a manager or superadmin from approving a file that the same person submitted. This is recorded as risk R-02 (Section 13).

4. Logical Architecture

4.1 Context

Figure 2 shows the platform, its users and the external services it uses. Browsers reach it over HTTPS. The server calls public data services for ship and aircraft positions, exchange rates and vehicle identification data. Continuous integration and deployment run in the source hosting service.

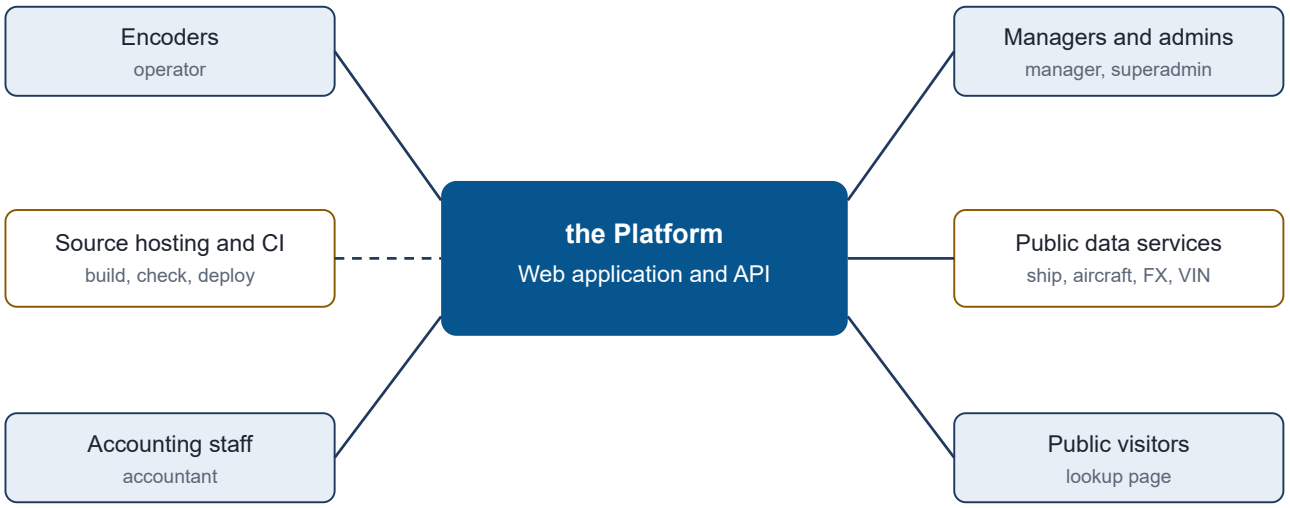


Figure 2. Context diagram. External systems are shown in amber.

4.2 Architectural style

- **Single-page web client** served as static files by the same server.
- **Layered API**: routes validate input and check roles; domain modules hold the rules; a data layer reads and writes the databases.
- **Pure rule modules** for the billing engine, shipment rating, invoice mathematics, bridge posting, operation status, air consolidation roll-up and tracking. They take inputs and return results, so the checks run without a database.
- **Generic CRUD factory** for simpler entities, with paging, company scoping and audit behaviour.
- **Hooks** on the shipment write path: operation-status rules, rating and billing sync run in a fixed order on the server.
- **Company scoping** on every operational row, set by the server from the signed-in user.

4.3 Component view

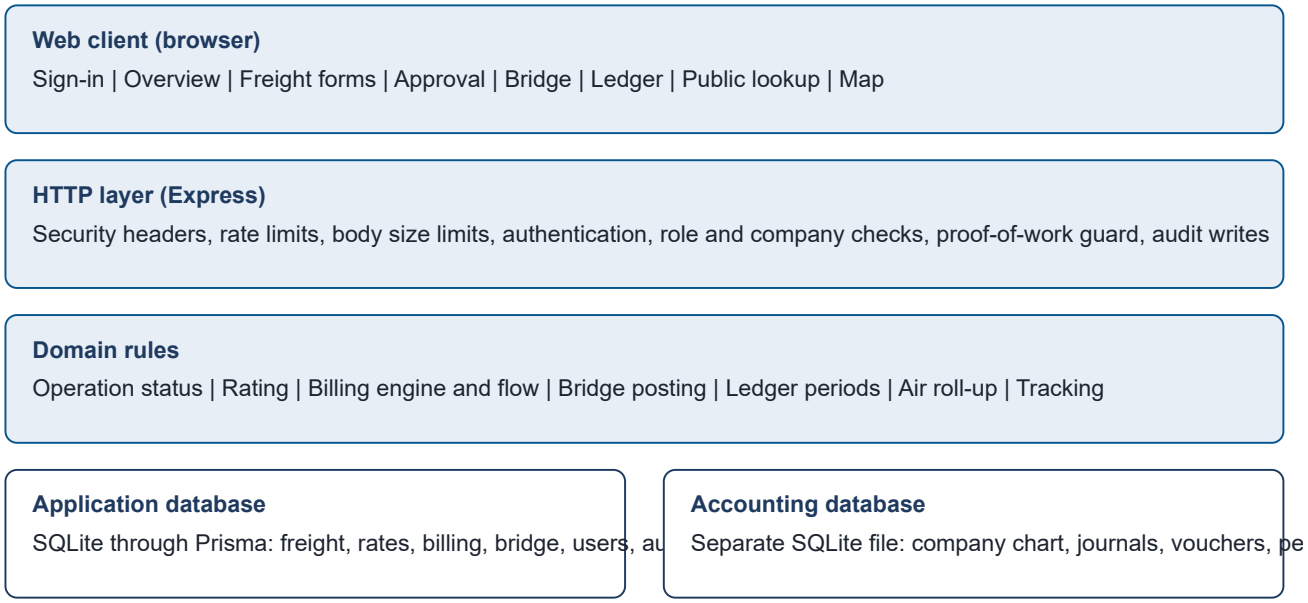


Figure 3. Component view. Each layer calls only the layer below it.

4.4 Layer responsibilities

Layer	Responsibility	Must not
Web client	Display, entry, validation feedback, calls the API with the access token, print and export in the browser	Hold secrets, decide access, post to the ledger
HTTP layer	Authenticate, authorise, validate with schemas, call domain rules, write audit records	Contain posting or rating rules inline
Domain rules	Approval transitions, charges, posting checks, period rules, consolidation roll-up	Read request objects or send responses
Data layer	Persist records, run start-up additive schema upgrades	Return rows across companies

5. Technology Stack

Table 5-1. Technology choices

Concern	Technology	Notes
Web framework	React 18 with Vite and TypeScript	Typed components; Vite builds the static bundle
Styling and UI	Tailwind CSS, Radix UI primitives, framer-motion, icon set, Inter typeface	Brand tokens; light and dark appearance
Client state and data	Zustand (persisted), TanStack React Query, React Hook Form with Zod	Session store and server data cache
Maps and charts	Leaflet with OpenStreetMap tiles, Recharts	Branch, port and route map with live feeds
Documents and export	jsPDF with autotable; spreadsheet export	Documents generated in the browser
API server	Node.js with Express and TypeScript	Run time Node.js 22 or later (node:sqlite); CI uses Node.js 24
Validation	Zod schemas on the server	Request bodies validated before rules run
Authentication	JSON Web Tokens (access and refresh), bcrypt password hashing	Lifetimes set by configuration
Security middleware	helmet, express-rate-limit, cookie-parser, cors	See Section 11
Application database	SQLite through Prisma 5	Single file
Accounting database	SQLite through node:sqlite	Separate file; path set by configuration
Hosting	Azure App Service (Linux), production slot	Serves API and web client from one process
Continuous integration	GitHub Actions	Checks on pull requests; deployment on push to main

6. Deployment and Operations

6.1 Deployment topology

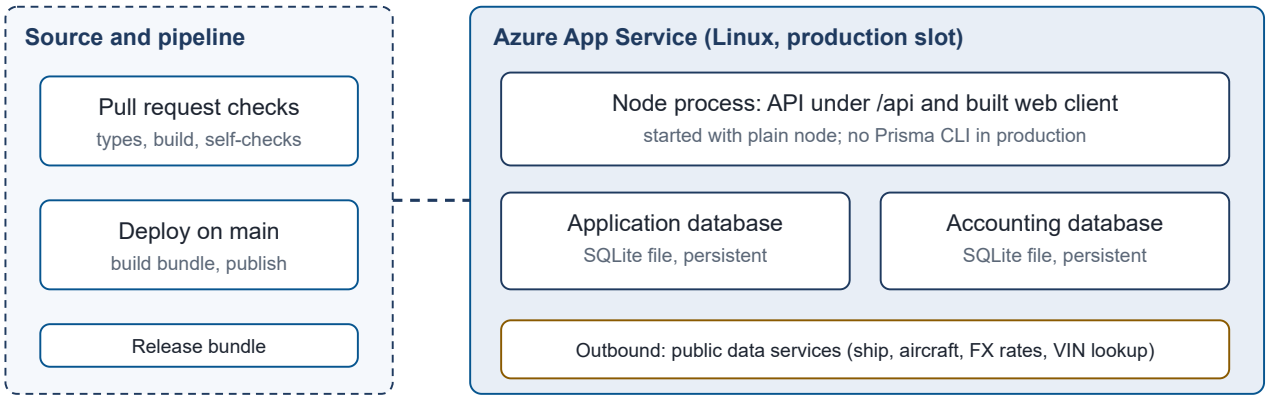


Figure 4. Deployment topology. The bundle holds the built web client, the compiled server, production dependencies and the schema files without data.

6.2 Release process

1. Work is developed on a branch. Each pull request runs the web checks (type check, build, export check) and the server checks listed in Table 12-1.
2. Changes merge to main after review and green checks.
3. A push to main runs the deployment workflow: install, build the web client, generate the database client, build the server, prune development dependencies, assemble the bundle and publish to the production slot. A concurrency group stops overlapping releases.
4. After a deploy, the health endpoint and the live site are checked before a demo or hand-over.

6.3 Configuration

Configuration comes from environment variables set in the host's application settings. The example file names the variables and states which are required in production. Values are never stored in the repository.

Table 6-1. Configuration variables (names only)

Variable	Purpose
PORT, CORS_ORIGIN	Listening port and allowed browser origins
DATABASE_URL, ACCOUNTING_DB_PATH	Locations of the application and accounting databases
JWT_ACCESS_SECRET, JWT_REFRESH_SECRET	Token signing secrets; required in production
ACCESS_TOKEN_TTL_MIN, REFRESH_TOKEN_TTL_DAYS	Token lifetimes
SEED_ADMIN_USERNAME, SEED_ADMIN_PASSWORD	Initial administrator; required in production
AIS_API_BASE, AIS_API_TOKEN	Ship position feed endpoint and optional token; server-side only
FLIGHT_API_BASE	Aircraft position feed endpoint; empty disables the feed
API_RATE_LIMIT_PER_MIN	Per-minute API request limit

Variable	Purpose
SMTP settings (Azure)	Arrival notice email sending; without them the send is a preview

6.4 Schema management

Production has no migration command. New tables and columns are added by a start-up routine that runs additive statements only; the Prisma schema file is kept in step. Start-up checks run before the server accepts requests. Destructive changes need a planned, reviewed data migration. A legacy ledger cleanup command exists in dry-run form and takes a verified backup before any change; it has been tested on synthetic ledgers only.

6.5 Operations

- **Health:** a health endpoint for hosting probes. **Built**
- **Logging:** request logging; audit records and booking events in the database. **Built**
- **Backup:** the legacy cleanup command makes a verified copy before changes; a routine backup and restore procedure for both databases is not in the repository. **Planned**
- **Demo copy:** a seeded copy of the databases is used for demonstrations, kept apart from production. **Built**
- **Maintenance:** a command clears operational records for test cycles; administrators only; never on production data.

7. Module Design

Each module lists its purpose, screens and routes, the records it owns, its rules and its build status. Route groups are listed in Section 10.

7.1 Operations overview

Purpose	Sign-in landing page: draggable overview with twelve widgets, a to-do list and a workspace per role.
Rules	Counts come from the company's live records. Income for the month is summed by estimated or actual arrival and departure dates, because no booking date exists in the data model (known limitation).
Status	**Built**

7.2 Ocean and air freight files, and approval

Purpose	Book and manage ocean and air files for export, import and domestic movements, with approval.
Records	Shipment (mode, direction, file number, parties, ports or airports, carrier, vessel or flight, containers, dimensions, master and house numbers, billing and cost lines, extra fields as JSON, operation status and its submit, review and return fields). Status events record each change with user, time, from and to status, note and optional image.
Approval rules	Operators, managers and superadmins create and edit while a file is draft or for correction; pending files are locked; only managers and superadmins approve or return; a return needs a note; approved files are locked for operators; accountants see approved files only; no generic write may change the approval fields.
Rating rules	Ocean revenue weight: the larger of gross tonnes and cubic metres. Air chargeable weight: the larger of gross kilograms and volumetric weight (length times width times height divided by 6000 per piece), rounded by the carrier rule (default up to the next half kilogram). Container and weight warnings are shown on the form.
Documents	Bill of lading, house bill of lading, air waybill, packing list, commercial invoice, manifest, export declaration and arrival notice are produced in the browser from the file. Branded layouts are checked in the test suite.
Form aids	Smart ports and airports by party country; currency list with USD default and correct decimals; commodity prefill; duplicate booking and master number checks.
Status	**Built**

7.3 Air consolidation

Purpose	One master air waybill (MAWB) carries many house air waybills (HAWB), each with its own shipper, consignee, pieces and weight.
Records	HAWB rows are stored with the MAWB file (extra data); each row has HAWB number, shipper, consignee, notify party, pieces, gross weight, dimensions and entered chargeable weight.
Rules	Pieces and gross weight roll up to the MAWB. The MAWB chargeable weight is the greater of total gross and total volumetric weight, not the sum of HAWB chargeable weights. Duplicate HAWB and MAWB numbers are flagged. The consolidation manifest, MAWB, HAWB and arrival notice read the rows.
Known limitation	The invoice house number for a consolidation repeats the MAWB number, because no single house number exists on a consolidation.

Status	**Built**

7.4 Vehicle inventory (roll-on/roll-off)

Purpose	Track each vehicle on a roll-on/roll-off booking from intake to shipment.
Records	Vehicle with identification number, year, make, model, colour, shipper, consignee, ports, booking number, location, condition and history events (JSON).
Rules	Status moves through Expected, Received, Ready to Ship, Pre-Loaded, Loaded, Shipped or Hold. Title status is Pending, Received or Missing. Customs hold and lien release are flags. A VIN can be decoded from a public service (US-format VINs only; the engine label shows engine layout, not displacement).
Status	**Built**

7.5 Pickup and delivery cartage

Purpose	Plan pickups, deliveries and cross-docks, and assign drivers and equipment.
Records	Cartage order with order number, type, status, addresses, schedule, cargo items (JSON), linked file and reference numbers.
Rules	Status is Open, Assigned, In Transit, Delivered or Cancelled.
Status	**Built**

7.6 Fleet, drivers and dispatch

Purpose	Maintain the fleet and driver registry; dispatch routes with proof of delivery.
Records	Fleet vehicle (Available, In Use, Maintenance); driver (Available, On Trip, Off Duty); dispatch route (Draft, Dispatched, In Transit, Arrived, Completed; stops as JSON; signature as data with signer and time).
Rules	A route references a driver, a vehicle and a cargo reference. Signature capture is limited to the connected signature-pad connector (connectors Planned).
Status	**Built** for registries and routes; **Partial** for hardware signature capture.

7.7 Rates, tariffs, charge packages, quotes and commodities

Purpose	Reference data that rating and documents depend on.
Records	Carriers; ports and airports (sea, air, inland); airlines; cargo warehouses; parties (customers, shippers, consignees, agents, truckers, consolidators); billing codes with default rate, ledger account and VAT flag; tariffs; charge packages with lines by incoterm and mode; quotes; commodities with HS code and dangerous-goods flag.
Rules	Most specific tariff wins: customer and lane, then lane, then any. Expired or future tariffs are ignored. A minimum charge raises the sell amount. A manual line with the same code as a tariff line takes precedence. An accepted quote replaces tariff sell lines.
Currency	Twenty-plus currencies with correct decimals; USD by default for international freight; PHP for the local invoice.
Status	**Built**. Tariff coverage is data entry by the business and is incomplete at this date (known gap).

7.8 Billing engine, invoices and billing flow

Purpose	Rate approved files into billing lines (sell), cost accruals (buy), VAT and invoices, and release them to the bridge.

Engine	A pure function takes the file and company tariffs and returns billing lines, cost lines, VAT, notes and rating messages. Engine lines carry an automatic flag and the tariff identifier; manual lines are not changed by the engine.
Flow	Runs on the server for every save of an air or ocean file, as the system, independent of the user's ledger access. Released only for approved files. One live invoice per file; a re-rate after issue marks the invoice for re-issue rather than changing it.
Invoices	Statuses draft, issued, posted and void. Reference numbers (mode code, date, daily sequence) at draft; invoice numbers (mode code, year and month, monthly sequence) at issue. VAT treatment 12 percent, zero or exempt. Invoice preview shows the FS journal before posting. Void creates a reversal voucher in the open period and the next revision is drafted from the file.
Automation modes	Company setting: manual (drafts only), review (default; drafts are staged for one-click posting), auto (posts when the period is open, vouchers balance and no rate note is open).
Limitations	No credit note document: an invoice with a zero or negative total is refused; correction is void and re-issue (known gap, Planned). All charges post to the sales account; the split by service awaits the accountants' decision (known gap, Planned).
Status	**Built** for the engine, flow, modes, invoices, preview, void and reversal. Credit note and revenue split are **Planned**.

7.9 Accounting bridge

Purpose	The only route from freight and billing into the ledger: stage, trial post, post.
Records	Bridge item with type (AR invoice, AP cost, miscellaneous invoice), status (Staged, Trial Verified, Posted), charge code, amount, source file and invoice, and an idempotency key. Check disbursement with status Draft, Printed, Posted or Void.
Rules	Each charge code maps to a ledger account; an unmapped code is refused with the reason. Trial posting builds the journal and checks that debits equal credits without writing. Final posting writes the balanced journal, sets Posted and records the journal reference. Voucher numbers are fixed per invoice (sales invoice, receipt, reversal) and checked before every write.
Queue	Stage funnel, drawer and tabs for operations queue, checks and disbursements, miscellaneous invoices and audit reports.
Status	**Built**

7.10 Accounting ledger (FS)

Purpose	General ledger for the company: vouchers, journals, chart of accounts, posting, period close and statements.
Company chart	The chart, report groups, subsidiary groups and the accounts that billing and the bridge post to are defined for the company code of this deployment only. Companies without a chart receive no ledger data. A versioned chart definition is applied by a dedicated command.
Records	Chart of accounts (account, sub-account, subsidiary); group codes; subsidiary groups; banks; suppliers; signatories; vouchers; journal entries (CDV, JE, AJE, AR, AP); transaction lines; fiscal periods. Stored in the accounting database with the ledger table prefix.
Rules	Journals balance. Entries dated in a closed month are refused with the first open date. The register shows the open period; closed months are reached through queries. Month-end closes the period and creates closing entries.
Reports	Trial balance, income statement, balance sheet, cash flow, check voucher report, subsidiary schedule; query browser; in-app accounting manual.
Legacy cleanup	A migration command finds leftover rows from other companies and test vouchers, reports conflicts, and removes only unreferenced rows in one transaction after a verified backup. It has been tested on synthetic ledgers and not yet run on production (known gap).

Status	**Built**; legacy cleanup run on production pending.

7.11 Tracking, public lookup and customer accounts

Purpose	Record milestones and let anyone with a reference see the status.
Records	Tracking item (reference, type Shipment, PD or Vehicle, status, linked file, milestones as JSON with status, location, note and time). Web account records hold a username, an email and permission flags.
Public lookup	Matches a file number, booking number, air or ocean waybill, master number or house number (including house rows of a consolidation). Each lookup needs a solved single-use proof-of-work token, and is rate limited. The response contains tracking fields only.
Customer sign-in	No customer sign-in or customer-scoped file view exists in code. **Planned**
Status	**Built** for staff tracking and the public lookup; **Planned** for the customer portal.

7.12 Live map, ships and aircraft

Purpose	Show branches, terminals, ports, active routes and ship and aircraft positions.
Rules	Ship positions come from a public feed and aircraft from a self-hosted feed, both called by the server. Queries are limited to a box of at most 20 degrees per side or a radius of 250 nautical miles; at most 1,500 markers are returned; a short in-memory cache applies. A missing feed token leaves ships empty; a missing aircraft endpoint reports "not configured".
Status	**Built**. Needs internet for map tiles and the ship feed token (known limitation).

7.13 Administration, audit, attachments, support, integrations and demo seed

Users and companies	Create, change and deactivate users; set roles and company access; password rules and history; superadmin protections. **Built**
Audit	Audit records for each mutating request; booking events for status changes. **Built**
Attachments	Files stored as encoded data on a record, up to 15 MB per request; used for return images. **Built**. A separate file store is **Planned**.
Support	Sign-in help requests create support tickets. **Built**
Integrations	Integration records hold status (disconnected, connected, error) for scanner, signature pad, label printer, government filing and carrier APIs. No live connector code exists. **Planned** for live connectors; status records **Built**.
Demo seed	Creates a demonstration set: twenty-two freight files across the four operation statuses, bridge states, parties, tariffs, charge packages, commodities, ports, carriers, tracking, RORO vehicles, cartage, quotes, web accounts and ledger entries that reconcile. Demo rows are marked and can be found. The seed must not run against production data. **Built**

8. Data Design

8.1 Data stores

The platform uses two SQLite databases. The application database holds freight, approval events, rates, billing, bridge, tracking, users, sessions, security events and audit records. The accounting database holds the ledger and is opened separately so its schema and backups are managed on their own.

8.2 Main entities and relationships

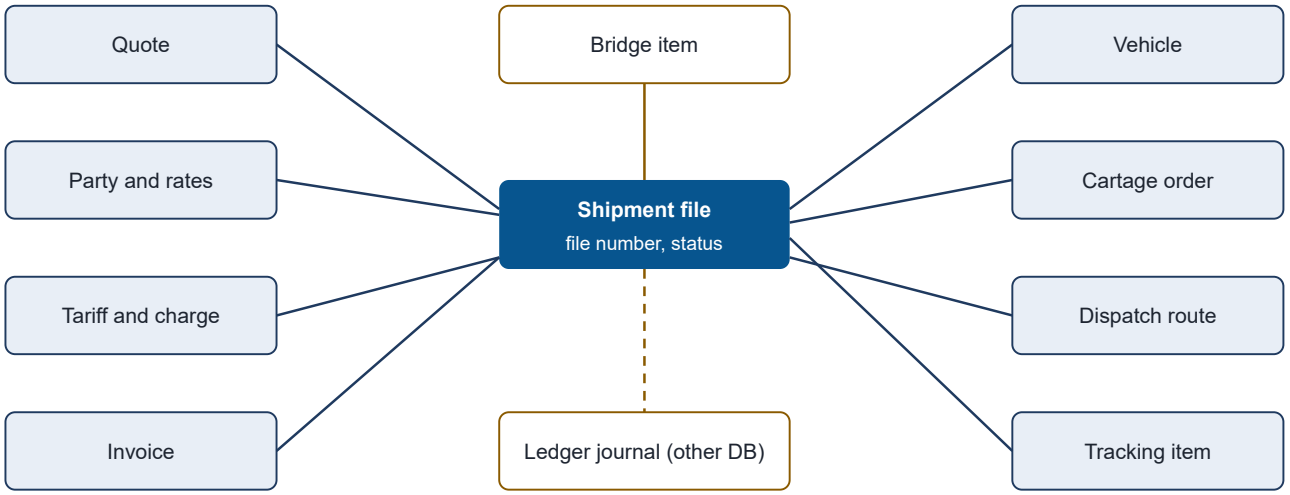


Figure 5. Entity overview. Lines show the link through the shipment file number and the bridge.

8.3 Entity catalogue

Table 8-1. Entities of the application database

Group	Entities	Purpose
Access	User, refresh token, password history, security event, support ticket	Sign-in, token rotation, password reuse, security trail, help requests
Tenancy	Company	Company profile and settings, including the billing-to-ledger mode
Freight	Shipment, operation status event, quote, commodity	Freight files with approval fields and lines; approval history; quotations; catalogue
Operations	Vehicle, cartage order, driver, fleet vehicle, dispatch route	Movement records
Rates	Carrier, port, airline, cargo warehouse, party, billing code, tariff, charge package and lines	Reference and rating data
Billing	Invoice, invoice sequence	Invoices and numbering counters
Bridge	Bridge item, check disbursement	Items for the ledger; check register
Tracking	Tracking item, web account	Milestones and customer access flags
Common	Attachment, integration, audit log	Files, connector status, change history

8.4 Data rules

- Each operational row has a company code set by the server from the signed-in user, never from the client.

- The approval fields (status, revision, submit and review fields, return note) are written only by the approval module.
- Unknown or empty approval values read as pending: locked and not approved. Legacy rows are approved because the column default was approved when added.
- Nested data (billing lines, cost lines, containers, dimensions, HAWB rows, milestones, stops) is stored as JSON text; rules that read them live in the domain modules.
- Money uses a dedicated money helper with exact decimal handling for invoice and ledger totals.
- Demo rows carry a demo marker so they can be identified and removed.

9. Key Process Sequences

9.1 Booking, submit, approve and bill

Table 9-1. Sequence: booking approved and billed

Step	From to	Action and result
1	Operator to web client	Enters a freight file; the form checks required fields and warns on weight, volume and duplicate numbers.
2	Web client to API	Create request with the access token. The shipment hook allows encoder roles and writes the file as draft under the user's company.
3	Operator to API (submit)	Submit checks the required fields. The file moves to pending, opRevision increases, and an event is written. The file is now locked.
4	Manager to API (approve)	Approve moves the file to approved and writes an event. Billing is released for the file.
5	Manager to API (return)	Return requires a note and may attach an image. The file moves to for correction; the operator edits and resubmits (back to pending).
6	API to billing engine and flow	For an approved file, rating creates billing and cost lines, syncs the invoice draft and stages items to the bridge in review mode.
7	Web client	Shows lines, rating messages, approval history and the billing stage.

9.2 Staged item verified and posted

Table 9-2. Sequence: bridge item trial posted and posted

Step	From to	Action and result
1	Billing flow to bridge	Staging creates bridge items with status Staged, keyed to prevent duplicates.
2	Accountant to web client	Opens the queue and selects items.
3	API to bridge rules	Each charge code is mapped to a ledger account. An unmapped code is refused with the reason.
4	Accountant (trial post)	Builds the journal in memory and checks debits equal credits. Status moves to Trial Verified; nothing is written to the ledger.
5	Accountant (post)	Writes the balanced journal, sets Posted, records the journal reference and writes an audit record.
6	Ledger	The journal appears in the register for the open period. A closed period refuses the entry.

9.3 Invoice void with reversal

An accountant voids a posted invoice. The system creates a reversal voucher in the open period that mirrors the posted vouchers, marks the invoice void, and drafts the next revision from the file. Posted data is never edited. A credit note document is not yet supported (Planned).

9.4 Sign-in with throttling and proof of work

Table 9-3. Sequence: sign-in

Step	From to	Action and result
1	Web client to API	Requests a signed proof-of-work challenge with an expiry.
2	Web client	Solves the challenge in a background worker and submits the credentials and the token.

Step	From to	Action and result
3	API	Verifies signature and expiry and refuses a replayed token.
4	API	Checks the password with bcrypt. Five failures start a one-minute block.
5	API	On success issues access and refresh tokens and records a security event.
6	Web client	Stores the session, attaches the token and refreshes it; inactivity ends the session.

9.5 Public lookup

A visitor enters a file, booking, waybill or house number. The browser obtains a single-use proof-of-work token; the server checks it, applies the rate limit, matches the number against files, consolidation house rows and tracking, and returns the status and milestones only.

9.6 Month-end close

The accountant runs month-end for the open period. The system creates the closing entries and marks the period closed. Later entries dated in that month are refused with the first open date.

10. Interface Design

10.1 Internal API

The API is served under /api. Every route except sign-in, challenge, public lookup and support requires a bearer token. Company and role checks run before each route. Lists support search and paging. Entity routes provide list, read, create, update and delete, subject to the role.

Table 10-1. Route groups

Path prefix	Owner	Use
/api/auth	Access	Sign-in, refresh, sign-out, current user, password change
/api/users, /api/companies	Administration	Users, companies and settings
/api/shipments (with operation status actions)	Freight	Files, submit, approve, return, status history
/api/quotes, /api/commodities	Freight	Quotations and commodity catalogue
/api/vehicles, /api/pd-orders, /api/drivers, /api/fleet-vehicles, /api/dispatch-routes	Operations	Vehicles, cartage, drivers, fleet, dispatch
/api/carriers, /api/ports, /api/billing-codes, master data routes	Rates	Reference directories
/api/billing	Billing	Settings, pipeline, queue, file flow, staging, invoices, charge packages, billing diagnostics
/api/bridge-items, /api/checks	Bridge	Bridge queue and check register
/api/tracking, /api/web-accounts	Tracking	Tracking records and web accounts
/api/attachments, /api/integrations, /api/audit-logs, /api/support	Common	Files, connector status, audit trail, support requests
/api/fs	Ledger	Vouchers, journals, chart of accounts, banks, suppliers, posting, month-end, reports, queries

10.2 Public endpoints

Table 10-2. Endpoints without a user sign-in

Endpoint group	Purpose and limits
Public lookup (file, booking, waybill, house and master numbers)	Read-only; proof-of-work token per lookup; rate limited; tracking fields only
Sign-in and challenge	Rate limited; throttled after repeated failures; proof-of-work
Support request	Creates a ticket; rate limited
Health and status	Service status for hosting probes

10.3 External data services

Table 10-3. External services used

Service	Used for	Notes
Open ship-position feed	Ship positions on the map	Called by the server; token optional and server-side
Aircraft position feed	Aircraft positions	Self-hosted; reports "not configured" when unset

Service	Used for	Notes
Exchange rate service	Live currency rates	Allowed for browser connections under the content security policy
Vehicle identification lookup	VIN decoding (US-format VINs)	Allowed for browser connections under the content security policy
Map tiles	Map background	Allowed for image loads
Email (SMTP)	Arrival notice sending	Preview only until configured in the host

10.4 Device and government connectors

Integration records hold the status of barcode scanners, signature pads, label printers, the government customs filing service and carrier APIs. No live connector code exists in the repository. Live connectors are **Planned**; features that need them stay disabled until the connector reports connected.

11. Security Design

11.1 Identity and sign-in

- Passwords are stored as bcrypt hashes; passwords over 72 bytes are refused; a password history supports reuse rules. **Built**
- Access tokens are short-lived and signed; refresh tokens are stored and revoked on sign-out. **Built**
- Five failed sign-ins start a one-minute block. **Built**
- Sign-in and public lookup each require a solved, signed, expiring, single-use proof-of-work token. **Built**. Password reset and registration do not exist in code.
- Inactivity ends the browser session. **Built**

11.2 Authorisation and approval controls

- Route guards check the role; viewers are refused every write.
- Ledger routes require a ledger access check; operators have none.
- Approval actions: encoders submit; managers and superadmins approve or return; a return requires a note; approved files are locked for operators.
- Accountants see approved files only, enforced in the query scope and not only in the screen.
- Approval fields cannot be set through a generic write. **Built**
- Company scope comes from the user's company list.
- Self-approval is not prevented (risk R-02).

11.3 Web hardening

- Security headers and a content security policy: scripts and fonts from the application only; images from map tile hosts and two public data hosts; browser connections only to the exchange rate and vehicle lookup hosts.
- Rate limits on sign-in, API, public lookup and support, with configurable per-minute API limit.
- Request body limits: 1 MB JSON, 100 KB form, 15 MB attachment route.
- Cross-origin access restricted to configured origins.

11.4 Data protection

- Public lookup responses carry tracking fields only.
- Feed credentials stay on the server.
- Secrets are not stored in the repository; the example file lists names only.
- Encryption of database files at rest depends on the host and is not configured by the application. **Planned** for review.

11.5 Audit

Each mutating request writes an audit record with user, company, action, entity and payload. Approval changes write booking events with from and to status, note and image. Ledger postings keep the journal reference.

11.6 Security review status

Self-checks cover user policy, HTTP behaviour, captcha and captcha enforcement, and run in continuous integration. A formal penetration test and dependency audit sign-off are **Planned**.

12. Testing and Quality Assurance

12.1 Automated checks in continuous integration

Table 12-1. Checks run on each pull request (server steps after build)

Check	Area
Web type check, build, export check	Web client
Server type check and build	Server
check:bridge	Bridge posting rules: balanced journals, mapping, refusals
check:billing, check:invoice, check:flow	Rating, invoice arithmetic and billing flow end to end
check:tracking, check:public-tracking	Milestones and public lookup
check:security, check:captcha, check:captcha-enforcement, check:validation	User policy, HTTP behaviour, captcha and validation
check:masterdata	Directory validation
check:benchmark	High-throughput concurrency benchmark
check:schema-drift, check:schema-heal	Schema matches the migration routine
check:air-booking, check:air-forms	Air booking fields and forms, consolidation roll-up
company chart check, qa:accounting	Company chart definition and accounting quality checks
check:opstatus	Operation status state machine and scopes

12.2 Quality assurance and measured results

The demo rehearsal on a copy of the databases measured: Overview first paint 0.45 to 0.6 seconds; opening a booking 0.3 to 0.4 seconds; saving 0.15 seconds; bridge queue with 200 files about 1.0 second (previously up to 2.1 seconds). The rehearsal reconciled the ledger: trial balance in balance on both sides, and sales revenue equal to the income statement. Load test notes are kept with the performance documents.

12.3 Test environments

- Local development runs the web client and the API on separate ports with synthetic data.
- Demonstrations run on a copy or a separate demonstration company, never on production records.
- End-to-end scripts exist; they are not part of the pull request checks. **Partial**
- Not covered by the rehearsal: tracking milestone entry screens, RORO inspection dialog, keyboard focus order and forced server errors (known gaps).

13. Risks, Known Gaps and Open Decisions

Table 13-1. Risk register (design view)

ID	Risk or issue	Rating	Mitigation or decision
R-01	Single-file databases limit concurrent writers; recovery depends on file copies	High	Routine backup and restore procedure (Planned); load tests show current limits
R-02	A manager or superadmin can submit and approve the same file; the code checks the role only	High	Decision needed: require a second approver or block self-approval
R-03	Billing automation depends on tariffs that are mostly not entered yet	High	Enter tariffs before relying on automatic billing; new lanes show "no tariff" until added
R-04	No credit note document; corrections use void and re-issue	Medium	Credit note Planned (known gap 1)
R-05	All charges post to the sales account; no split by service	Medium	Revenue split waits for the accountants' decision (known gap 2); Planned
R-06	Legacy ledger rows and the cleanup command have not been applied to production	Medium	Dry run first, verified backup, then one reviewed run (known gap 3)
R-07	Consolidation invoice house number repeats the MAWB number	Low	Known gap 8; accepted until a house number rule is set
R-08	Operators can edit each other's draft files (shared desk)	Low	Known gap 7; accepted; approved and pending files are locked
R-09	Live map and VIN decoding depend on internet and public services	Medium	Known gaps 6 and 9; the page states when a feed is not configured
R-10	Arrival notice email is a preview until SMTP is configured	Medium	Known gap 10; configure SMTP in the host before sending
R-11	Files created through the API get a generic file number, not the form prefix	Low	Known gap 11; use the form or add prefix rules
R-12	No live device, government or carrier connectors; no customer sign-in	Medium	Planned; manual entry continues
R-13	Credentials used during development must be rotated before go-live	Medium	Owner action: rotate the administrator password and feed tokens; set secrets in host settings only

Open decisions

- Self-approval rule (R-02): business and technical decision.
- Revenue split by service and the credit note design (R-04, R-05): accounting decision.
- Timing of the legacy ledger cleanup on production (R-06): owner decision after dry run.
- Removal of the stopped earlier production instance: owner decision.

14. Assumptions and Constraints

14.1 Assumptions

- Users have a modern browser with JavaScript enabled and a stable connection.
- Tariffs and charge packages are maintained by the accounting team and kept up to date.
- The hosting platform provides HTTPS, persistent storage for the database files and application settings for secrets.
- The company's chart of accounts matches the accounts that billing codes map to.

14.2 Constraints

- SQLite is the engine for both databases; the design assumes moderate concurrency.
- Production starts with a plain Node process; schema changes are additive and applied at start-up.
- Node.js 22 or later is required at run time for the accounting database module.
- Public feeds are used under their own terms, with attribution where required.

15. Design Traceability

The table links each design area to the requirements in the companion SRS.

Design area (SAD section)	Requirement groups (SRS)	Verification
Sign-in, proof of work and sessions (9.4, 11.1)	FR-AU-01 to FR-AU-08; NFR-SE-01 to NFR-SE-04	check:security, check:captcha, check:captcha-enforcement
Roles, company scope, approval controls (2.6, 3.3, 11.2)	FR-AU-09 to FR-AU-12; NFR-SE-05; FR-AP-01 to FR-AP-06	check:security, check:opstatus, route tests
Operations overview (7.1)	FR-OV-01 to FR-OV-02	Screen review
Freight files and rating (7.2)	FR-SH-01 to FR-SH-09	check:air-booking, check:invoice, check:billing
Air consolidation (7.3)	FR-SH-10 to FR-SH-12	check:air-forms, check:air-booking
Rates, tariffs and commodities (7.7)	FR-RT-01 to FR-RT-05	check:masterdata, check:billing
Billing and invoices (7.8, 9.3)	FR-BL-01 to FR-BL-08	check:billing, check:invoice, check:flow
Vehicles, cartage, dispatch (7.4 to 7.6)	FR-VH-01 to FR-VH-03; FR-PD-01 to FR-PD-03; FR-FD-01 to FR-FD-04	Functional review by module
Accounting bridge (7.9, 9.2)	FR-BR-01 to FR-BR-07	check:bridge, qa:accounting
Ledger and period close (7.10, 9.6)	FR-FS-01 to FR-FS-09	company chart check, qa:accounting
Tracking, public lookup, customer accounts (7.11, 9.5)	FR-TR-01 to FR-TR-05; FR-PT-01 to FR-PT-03	check:tracking, check:public-tracking
Live map and feeds (7.12, 10.3)	FR-MP-01 to FR-MP-03	check:tracking; screen review with feed on and off
Audit, attachments, support, integrations, demo seed (7.13)	FR-AD-01 to FR-AD-06	Audit review; route tests; demo rehearsal
Performance and availability (5, 12.2)	NFR-PF-01 to NFR-PF-03; NFR-AV-01	check:benchmark; rehearsal measurements; health check
Deployment and schema (6)	NFR-MT-01 to NFR-MT-03	check:schema-drift; deployment runs

16. Glossary

Term	Meaning in this document
AP / AR	Accounts payable (costs to suppliers) / accounts receivable (charges to customers)
AWB / HAWB / MAWB	Air waybill; house air waybill; master air waybill
Arrival notice	Document sent to the consignee on arrival of a consolidation or import, one per house
Bridge item	A billing or cost line staged for verification and posting to the ledger
BL / HBL / MBL	Bill of lading; house bill of lading; master bill of lading
CDV	Check disbursement voucher
Charge package	A set of charge lines applied by mode, direction and incoterm
Chargeable weight	The greater of actual and volumetric weight used to rate an air file
Consolidation	Several house air waybills under one master air waybill
DG	Dangerous goods
FCL / LCL	Full container load; less than container load
FS	The accounting ledger module
HS code	Harmonized System code for classifying commodities
Incoterm	Standard trade term defining who pays which charges
Journal	A balanced set of debit and credit lines in the ledger
Operation status	Approval state of a file: draft, pending, for correction or approved
Milestone	A timed status update on a tracking item
Reversal voucher	A ledger voucher that undoes a posted voucher when an invoice is voided
P/D	Pickup and delivery (cartage)
POD	Proof of delivery captured as a signature on a dispatch route
RORO	Roll-on/roll-off vehicle shipping
TACT	Air freight weight rounding convention; the default carrier rule here
Tariff	A rate for a charge code, optionally limited to a customer, lane and validity period
Trial post	A check that builds a journal and confirms it balances without writing it
VAT	Value-added tax; 12 percent on taxable billing lines
VGM	Verified gross mass of a container
W/M	Weight or measurement: the greater of weight and volume used for ocean revenue